Exercises 6-10

Exercise 8

In the lecture we learned the autoregression model $x_t = a_0 + a_1 x_{t-1} + \ldots + a_p x_{t-p} + w_t$

1. Write an R function “GenAR” which simulates autoregression model with Gaussian white noise. You need to achieve the following:

- The input of the function must include: length of simulated time series (sample size), variance of Gaussian white noise, the coefficient vector $(a_0, a_1, \ldots, a_p)$ and the initial value vector $(x_{-p+1}, x_{-p+2}, \ldots, x_0)$.
- The output is a vector (preferably a *ts* object) containing the simulated time series.

You may directly make use of “filter” function in R.

2. Use the function to simulate the following model

$$x_t - 2x_{t-1} + x_{t-2} = w_t, \quad t = 1, 2, 3, \ldots, 1000$$

where w_t is Gaussian white noise with variance equal to 2 , and initial values are set as $x_{-1} = 1, x_0 = 0$.

3. Discuss briefly what you observe. Does the time series curve look smoother or rougher compared to the random walk time series you saw in lecture?
4. Difference (with lag 1) the simulated time series and plot the resulting time series. What do you observe this time?

Part 1

```
# n is Sample size.
# w is Gaussian white noise
# sigma_squared is variance of Gaussian White Noise
# a is Coefficient vector (a0, a1, ..., ap)
# x_init is Initial value vector (x_{-p+1}, ..., x_0)
GenAR <- function(n, sigma_squared, a, x_init) {
  # We separate a0 (intercept) from autoregressive coefficients
  a0 <- a[1]
  phi <- a[-1]
  p <- length(phi)

  # We generate white noise w_t ~ N(0, sigma_squared)
  # If a0 is non-zero, it is added to the noise mean
  w <- rnorm(n, mean = a0, sd = sqrt(sigma_squared))

  # Recursive filtering: x_t = phi1*x_{t-1} + ... + phip*x_{t-p} + w_t
  # We need to reverse x_init's time order: (x_0, x_{-1}, ..., x_{-p+1})
  sim_data <- filter(w, filter = phi, method = "recursive", init = rev(x_init))

  # Return as a ts object
  return(as.ts(sim_data))
}
```

Part 2

Here, the given model is

$$x_t - 2x_{t-1} + x_{t-2} = w_t, \quad t = 1, 2, 3, \ldots, 1000$$

, which when we rearrange to its standard form, we get,

$$x_t = 2x_{t-1} - x_{t-2} + w_t$$

, where

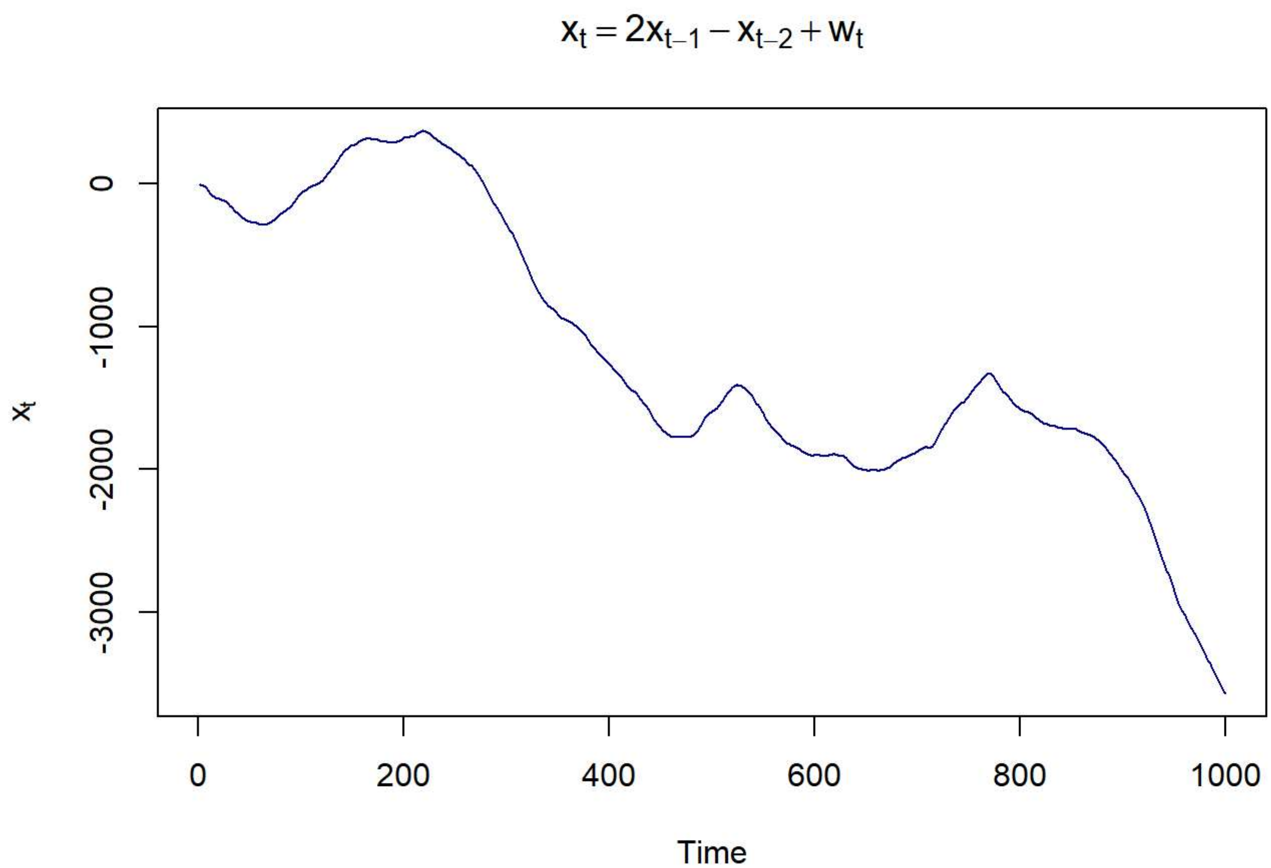
$$a_0 = 0, a_1 = 2, a_2 = -1$$

.

set.seed(21)

```
n <- 1000
sigma_squared <- 2
coeffs <- c(0, 2, -1) # a0, a1, a2
initial_vals <- c(1, 0) # x_{-1}, x_0

x_sim <- GenAR(n, sigma_squared, coeffs, initial_vals)
plot(x_sim, main = expression(x[t] == 2*x[t-1] - x[t-2] + w[t]),
     ylab = expression(x[t]), col = "darkblue")
```



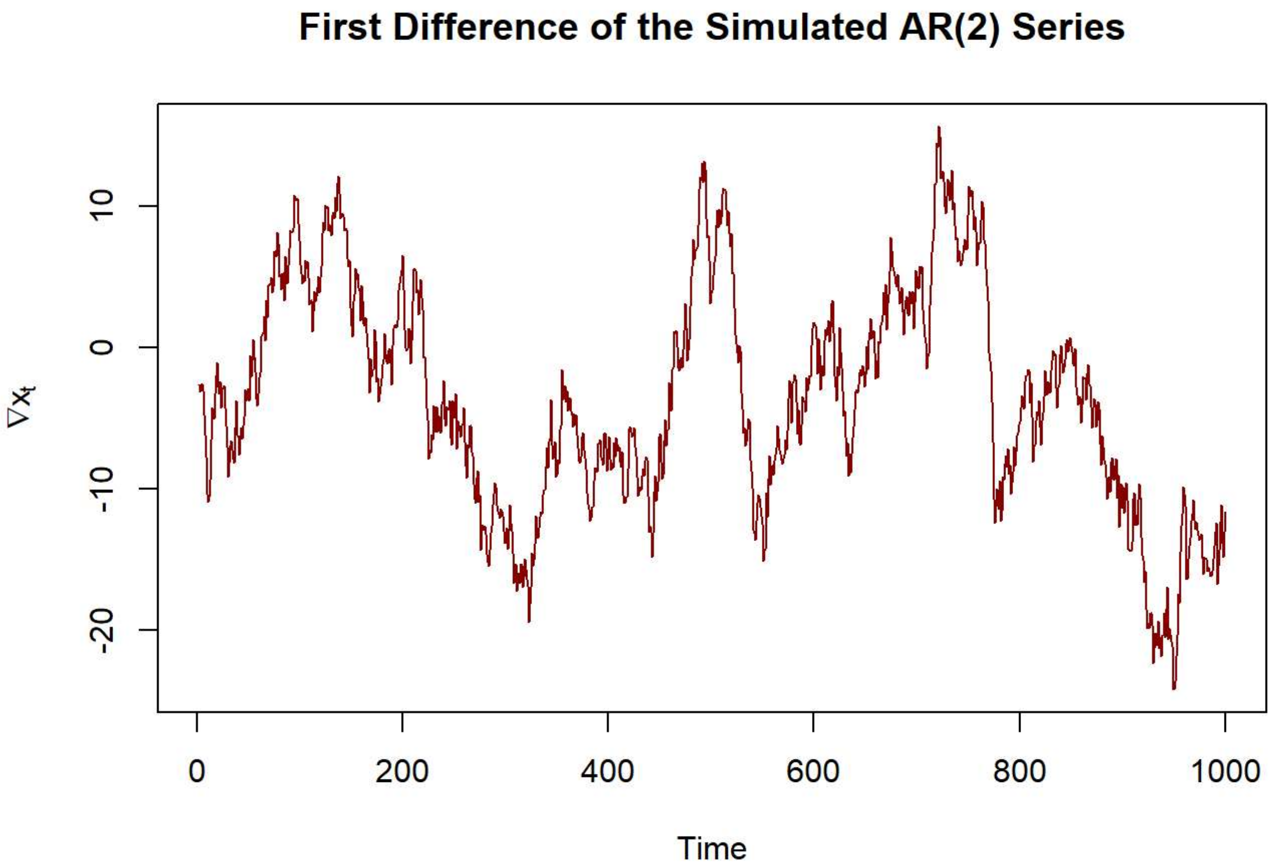
Part 3

We see a time series that is much smoother, and the trajectory is random, but smooth. This curve is at two levels of integration.

Part 4

When we difference it by lag 1, it removes one level of integration.

```
x_sim_2 <- diff(x_sim, lag = 1)
plot(x_sim_2, main = "First Difference of the Simulated AR(2) Series",
     ylab = expression(nabla*x[t]), col = "darkred")
```



We see that this is far less smooth, and one more level of differencing would yield a plot very similar to random walk simulations.

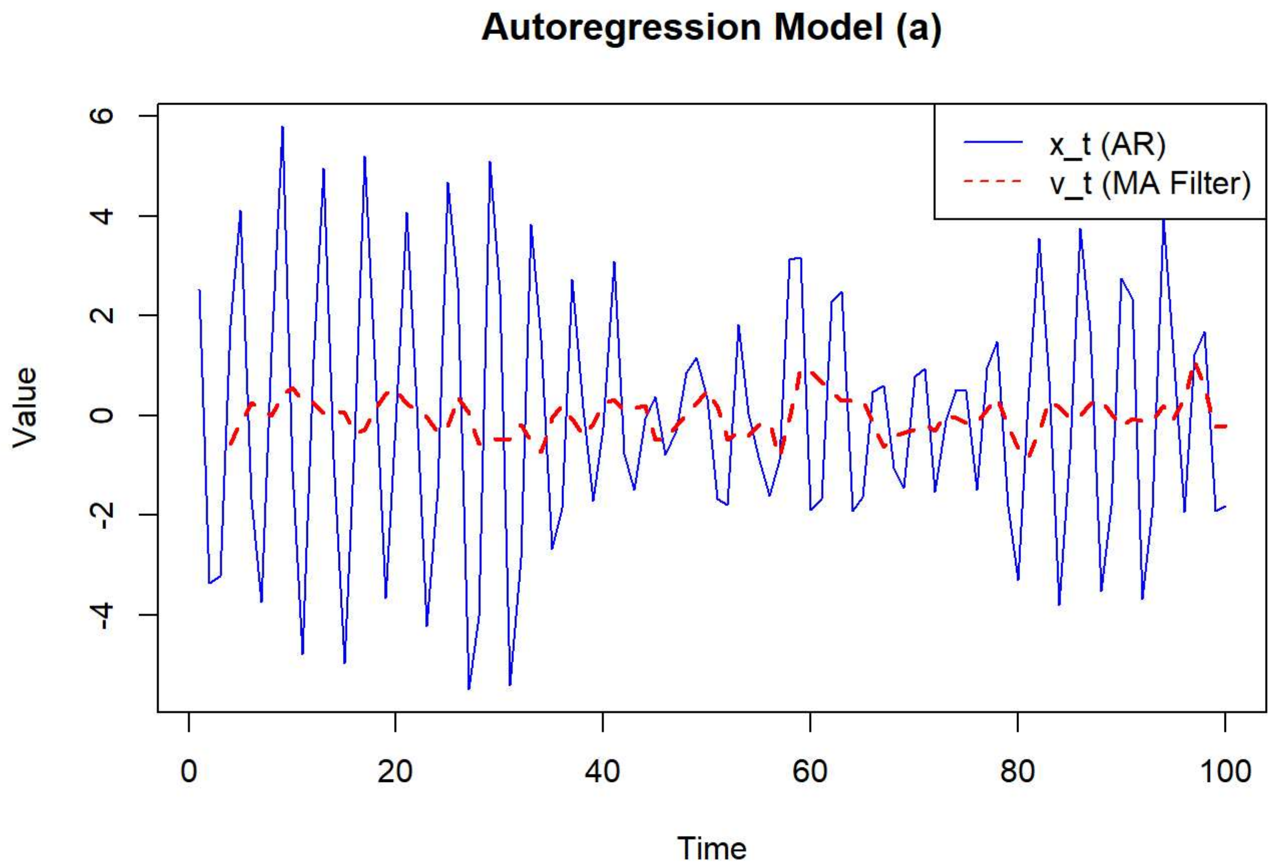
Exercise 9

a. In this part, we generate n = 100 observations from the autoregressive model x_t = -0.9 x_{t-2} + w_t and we apply a 4 point moving average filter.

```
# Using steps given in Example 1.10
w = rnorm(150, 0, 1) # Generating extra noise to handle startup
# AR(2) model: x_t = 0*x_{t-1} - 0.9*x_{t-2} + w_t
x = filter(w, filter=c(0, -0.9), method="recursive")[-(1:50)] # Remove startup
x = as.ts(x)

# Applying moving average filter: v_t = (x_t + x_{t-1} + x_{t-2} + x_{t-3})/4
v = filter(x, rep(1/4, 4), sides=1)

plot.ts(x, main="Autoregression Model (a)", ylab="Value", col="blue")
lines(v, lty="dashed", col="red", lwd=2)
legend("topright", legend=c("x_t (AR)", "v_t (MA Filter)"),
       col=c("blue", "red"), lty=c(1,2))
```

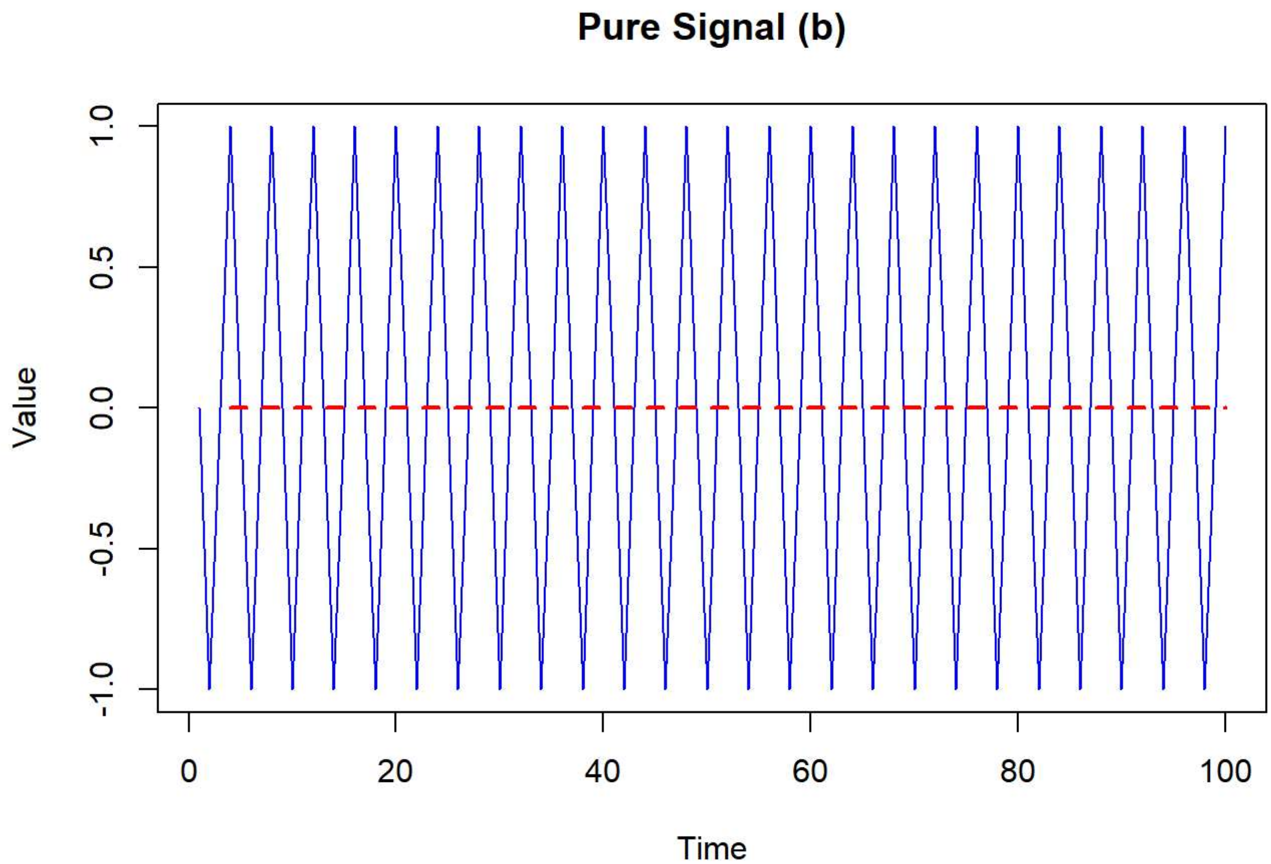


We see that x_t exhibits a quasi-periodic, oscillating behavior. When we apply the v_t moving average filter, we see that it smooths the series significantly, revealing a lower frequency periodic trend.

b.

```
t = 1:100
x_b = ts(cos(2*pi*t/4))
v_b = filter(x_b, rep(1/4, 4), sides=1)

plot.ts(x_b, main="Pure Signal (b)", ylab="Value", col="blue")
lines(v_b, lty="dashed", col="red", lwd=2)
```

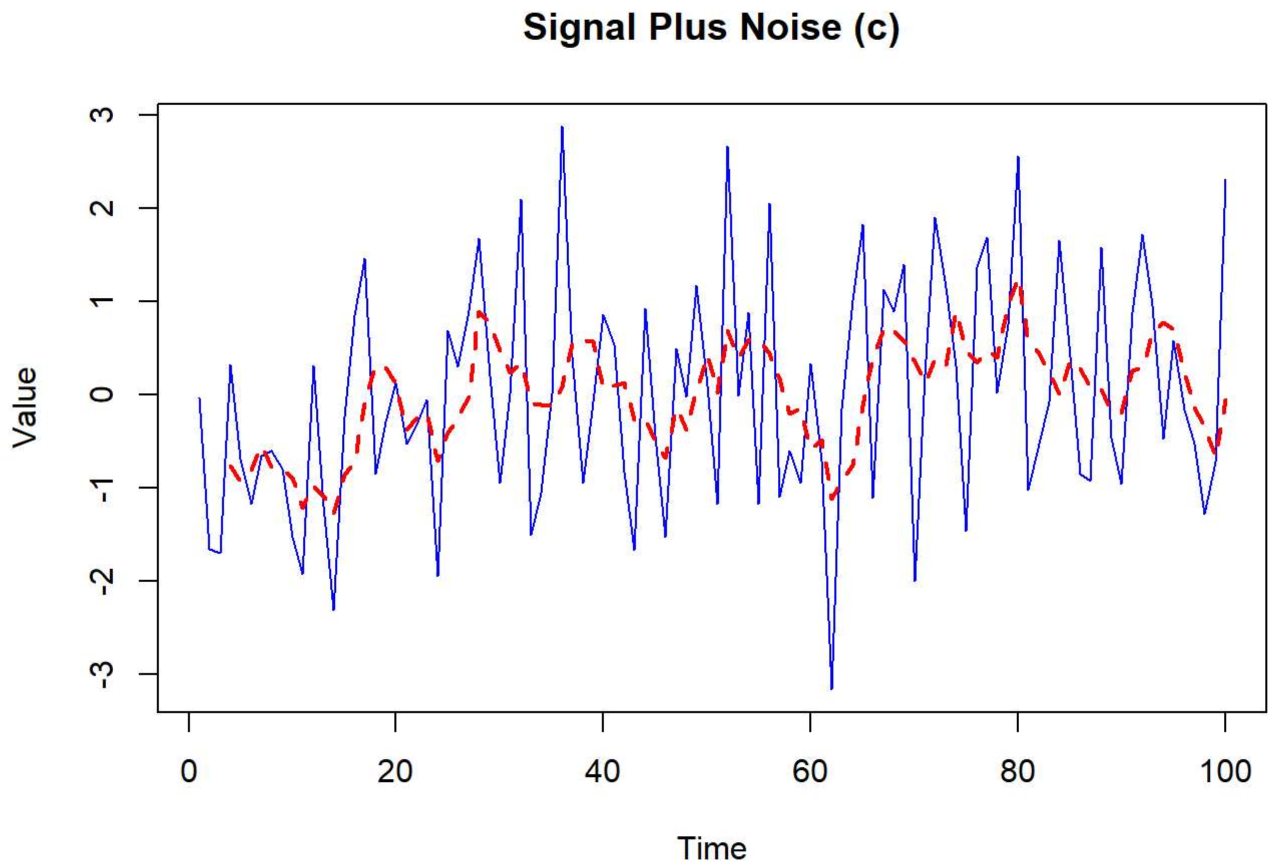


We see that the signal completes one full oscillation every 4 timesteps, and using a length of 4 leads to the entire cycle being averaged to the mean (0), and we get a flatline at zero.

c.

```
w_c = rnorm(100, 0, 1)
x_c = ts(cos(2*pi*t/4) + w_c)
v_c = filter(x_c, rep(1/4, 4), sides=1)

plot.ts(x_c, main="Signal Plus Noise (c)", ylab="Value", col="blue")
lines(v_c, lty="dashed", col="red", lwd=2)
```



After adding noise, we see that the signal retains the oscillatory trend, but is not uniform anymore. When we filter the wave, we see it gets smoother, and the cosine wave trend is much harder to discern, effectively filtering out the known signal entirely.

d. We see the following trends:

- a. represents an the autoregression function, and the filtering smooths out the plot, but retains the general path of the original function.
- b. represents a cosine function, on which the filter removes all information of the cosine curve into a flatline.
- c. noise modifies the cosine curve, on which, the filter retains some of the cosine features, albeit hard to discern.

Exercise 10

We need to show that:

$$\gamma(s,t) = E[(x_s - \mu_s)(x_t - \mu_t)] = E(x_s x_t) - \mu_s \mu_t$$

where $E[x_t] = \mu_t$.

First, we multiply the two terms $(x_s - \mu_s)$ and $(x_t - \mu_t)$:

$$(x_s - \mu_s)(x_t - \mu_t) = x_s x_t - x_s \mu_t - x_t \mu_s + \mu_s \mu_t$$

We apply expected operator to both sides:

$$E[(x_s - \mu_s)(x_t - \mu_t)] = E[x_s x_t] - E[x_s \mu_t] - E[x_t \mu_s] + E[\mu_s \mu_t]$$

Since μ_t = $E[x_s]$ and $\mu_s = E[x_s]$, and both are constant. Therefore:

$$E[x_s \mu_t] = \mu_t E[x_s] = \mu_t \mu_s$$

$$E[x_t \mu_s] = \mu_s E[x_t] = \mu_s \mu_t$$

$$E[\mu_s \mu_t] = \mu_s \mu_t$$

When we substitute these, we get:

$$\gamma(s,t) = E[x_s x_t] - \mu_s \mu_t - \mu_s \mu_t + \mu_s \mu_t$$

$$\gamma(s,t) = E[x_s x_t] - \mu_s \mu_t$$

This confirms that the autocovariance is the expectation of the product minus the product of the means.

Exercise 11

To find the autocorrelation function (ACF) of x_t, we first need to determine its mean and autocovariance function. Since s_t is nonrandom and $E[v_t] = 0$:

$$\mu_x(t) = E[x_t] = E[s_t + v_t] = E[s_t] + E[v_t] = s_t + 0 = s_t$$

The autocovariance $\gamma_x(s,t)$ = $E[(x_s - \mu_x(s))(x_t - \mu_x(t))]$.

Substituting our expressions for x_t and $\mu_x(t)$:

$$\begin{aligned} \gamma_x(s,t) &= E[(s_s + v_s - s_s)(s_t + v_t - s_t)] \\ &= E[v_s v_t] \end{aligned}$$

As we know, v_t has a mean of zero, the term $E[v_s v_t]$ is exactly the autocovariance of the noise series, $\gamma_v(s,t)$. Therefore:

$$\gamma_x(s,t) = \gamma_v(s,t)$$

The ACF is the normalized autocovariance:

$$\rho_x(s,t) = \frac{\gamma_x(s,t)}{\sqrt{\gamma_x(s,s)\gamma_x(t,t)}}$$

Substituting γ_x with γ_v:

$$\rho_x(s,t) = \frac{\gamma_v(s,t)}{\sqrt{\gamma_v(s,s)\gamma_v(t,t)}} = \rho_v(s,t)$$

Hence, we can conclude that the autocorrelation function of the signal-plus-noise series is identical to the autocorrelation function of the noise series:

$$\rho_x(s,t) = \rho_v(s,t)$$